Sandeep Bhupatiraju

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PROFESSIONAL SUMMARY

AI Solutions Architect with 10+ years of experience spanning advanced research and production systems development. Over the past 6 years, built production LLM systems, agentic workflows, and RAG platforms for governance and legal analytics, including **100M+ legal record databases** across multiple countries. Developed AI applications using LangGraph, CrewAI, Neo4j, and Azure AI Search. **Ph.D. in Mathematics (Probability Theory)** with peer-reviewed research on bias detection and responsible AI in judicial systems. Expertise in multi-agent orchestration, knowledge graphs, enterprise AI integration, and responsible AI governance for policy applications.

TECHNICAL EXPERTISE

Agentic AI & Large Language Models

- Multi-Agent Orchestration: LangGraph, LangChain, CrewAI, Atomic Agents for autonomous workflows
- Design Patterns: Reflection, consensus mechanisms (judge-advocate-skeptic), entity resolution
- LLM Platforms: OpenAI API, AWS Bedrock (multimodal models for OCR and structured outputs)
- Prompt Engineering: DSPy for automated optimization, structured outputs via Pydantic, custom evaluation frameworks

RAG & Information Retrieval

- Vector Search: Azure AI Search, Pinecone, Chroma, FAISS
- Advanced RAG Techniques: Graph RAG with Neo4j, HyDE, query decomposition, cross-encoder reranking
- Embeddings & Optimization: Sentence transformers, OpenAI embeddings, retrieval quality evaluation

Data Engineering & Cloud Infrastructure

- Data Pipelines: Databricks, Delta Live Tables (DLT), Medallion architecture (Bronze/Silver/Gold)
- Databases: Neo4j (Cypher), PostgreSQL, MongoDB, vector databases
- Cloud & Deployment: AWS (Bedrock, Lambda, S3), Azure (AI Search, OpenAI, Functions), containerized deployment (Docker, Heroku), Fly.io
- Integration: Event-driven architecture (Lambda triggers), RESTful APIs (OpenAPI), CI/CD pipelines

ML/AI Operations & Domain Expertise

- Monitoring & Observability: LangSmith for trace analysis, latency tracking, automated error detection
- Evaluation: Custom eval frameworks, A/B testing, performance metrics, human-in-the-loop validation
- AI Governance & Safety: Input validation, output filtering, guardrails for production LLM systems
- Model Optimization: LoRA (parameter-efficient fine-tuning), knowledge distillation, quantization for local LLM deployment
- Policy Analytics: Judicial bias detection, governance inefficiencies, technology adoption analysis
- Knowledge Systems: Expertise quantification, entity linkage, knowledge graph construction

Programming & Tools

- Languages: Python (NumPy, TensorFlow, PyTorch, Keras, Pydantic, Scrapy), MATLAB, Mathematica, Cypher
- Tools: Git, Docker, Neo4j, Graph visualization (Gephi, Neo4j Bloom)

PROFESSIONAL EXPERIENCE

AI Consultant (AI Solutions Architect) 2024–2025

Paradigm Case Management, Denver, Colorado

Led AI architecture and engineering for legal workflow automation platform serving prosecutors and case management professionals. Designed and deployed production agentic systems using LangGraph, CrewAI, and Neo4j knowledge graphs.

Multi-Agent Systems & Orchestration

- Architected LangGraph-based orchestration pipelines for structured information extraction, entity resolution, database record updates, and verification workflows
- Designed multi-agent entity resolution system identifying identical persons across heterogeneous document types (court filings, police reports, case records)
- Implemented reflection pattern for iterative accuracy improvement; built consensus-based extraction using judge-advocate-skeptic pattern for ambiguous legal definitions

Knowledge Graphs & Graph RAG

- Developed Graph RAG system over Neo4j connecting case-level information using graph transformers
- Implemented Cypher queries for complex relationship traversal; designed knowledge graph schema optimized for legal entity relationships and temporal case evolution

Workflow Automation & Integration

- Built event-driven workflows using AWS Lambda triggers for automated document processing (summarization, record linkage, missing document identification)
- Deployed multimodal LLMs from AWS Bedrock for OCR and structured output extraction from scanned legal documents
- Developed dynamic Pydantic response models using FAISS for efficient enum generation
- Led technical implementation with engineering team; conducted code reviews and established engineering best practices for AI pipeline development; coordinated with DevOps teams; managed junior technical staff

Technologies: LangGraph, LangChain, CrewAI, Atomic Agents, Neo4j, AWS (Lambda, Bedrock, S3), Python, Pydantic, FAISS

Consultant (AI & Data Analytics) 2018–Present

World Bank Group, Washington D.C.

Lead AI solutions development and large-scale data engineering for governance analytics across multi-country projects. Built production RAG systems, expertise discovery platforms, and datasets for policy research.

AI & Machine Learning Projects

- **K360 Knowledge Platform (RAG System):** Designed enterprise retrieval-augmented generation system using Azure AI Search for multi-document question answering; implemented query decomposition, cross-encoder reranking, synthesis optimization; collaborated on OpenAPI integration and architecture decisions
- **Sherlock Expertise Discovery System:** Architected novel expertise quantification system aggregating signals across heterogeneous data sources; designed retrieval algorithms for subject matter expert matching; formulated multi-dimensional expertise metrics balancing recency, depth, and breadth
- **Enterprise Surveys Citation Analysis:** Developed quantitative methodology to identify and measure Enterprise Surveys usage in World Bank reports; analyzed citation patterns to understand how policymakers use survey data in country engagement processes; identified frequently cited survey components
- **GEF Technology Adoption Analytics:** Identified systematic technology adoption patterns across Global Environment Facility project portfolio; quantified gaps between advisory recommendations for technology use and actual implementation in projects; provided evidence-based insights for improving technology integration
- **GEF Policy Coherence Assessment:** Analyzed policy coherence within GEF's project portfolio; developed framework for measuring alignment between policy recommendations and project implementation
- **Municipal Governance Analytics (Lithuania & Estonia):** Compiled and analyzed municipal meeting transcripts across all municipalities in Lithuania and Estonia; applied multilingual NLP and topic modeling to quantify discussion topics; tracked public participation patterns and policy emphasis shifts across countries and over time

Technical Leadership & Collaboration

- Led technical reviews and mentored developers and data scientists on RAG implementation, API design, and embedding strategies; provided architectural guidance for knowledge platform evolution
- Coordinated with product managers, data scientists, and engineering teams across time zones to deliver AI solutions aligned with organizational knowledge management goals
- Provided technical guidance on embedding models, retrieval strategies, and evaluation frameworks

Data Engineering & ETL

- Built large-scale legal datasets: India (83M district court, 8M high court, 150K appellate cases), Indonesia (12M records), BOOST budget data (multi-country public finance datasets)
- Designed ETL pipelines using Databricks and Delta Live Tables with Medallion architecture (Bronze/Silver/Gold)
- Developed web scraping and data extraction pipelines (Scrapy, custom Python tools)

Policy Analytics & Research

- Produced research on judicial bias (gender, religion, caste), environmental justice litigation, technology adoption
- Applied supervised ML models and LLM pipelines for classification, bias detection, policy evaluation
- Supervised and mentored research assistants and interns across India, Kenya, Philippines

Technologies: Azure AI Search, Databricks, Delta Live Tables, PostgreSQL, OpenAPI, sentence transformers, cross-encoders, Python

Mentor 2019–2020
World Bank Intern Program, Washington D.C.
Supervised intern cohorts working on data extraction and analysis projects from India, Kenya, and the Philippines; mentored interns on advanced NLP techniques, model training, and evaluation methodologies.

Associate Instructor 2010–2018
Indiana University, Bloomington, USA
Taught undergraduate mathematics courses including Calculus, Linear Algebra, Mathematical Statistics, and Probability Theory.

EDUCATION

Ph.D. in Mathematics (Probability Theory), Indiana University, Bloomington, USA 2019
Dissertation: Low-intensity Limits of Random Voronoi Tessellations

M.S. in Mathematics, Indian Institute of Science, Bangalore, India 2010

B.Sc. in Mathematics, Loyola College, University of Madras, Chennai, India 2007

PUBLICATIONS

Peer-Reviewed Journal Articles

- Ash, E., Asher, S., Bhowmick, A., **Bhupatiraju, S.**, Chen, D. L., Devi, T., Goessmann, C., Novosad, P., & Siddiqi, B. (2025). Measuring gender and religious bias in the Indian judiciary. *Review of Economics and Statistics*, 1–45.
- **Bhupatiraju, S.**, Chen, D., & Venkataramanan, K. (2024). Mapping the Geometry of Law Using Natural Language Processing. *European Journal of Empirical Legal Studies*, 1(1), 49–68.
- Chen, D. L., Joshi, S., **Bhupatiraju, S.**, & Neis, P. (2024). Caste aside? Names, Networks and Justice in the courts of Bihar, India. *European Journal of Empirical Legal Studies*, 1(2), 151–178.
- **Bhupatiraju, S.**, Chen, D. L., Joshi, S. (2021). The promise of machine learning for the courts of India. *National Law School of India Review*, 33, 463.
- **Bhupatiraju, S.**, Hanson, J., & Járai, A. A. (2017). Inequalities for critical exponents in d-dimensional sandpiles. *Electronic Journal of Probability*.

Book Chapters

- **Bhupatiraju, S.**, Chen, D. L., & Joshi, S. (Forthcoming). The Promise of AI for the Courts in India. In *The Cambridge Handbook of AI and Technologies in Courts*. Cambridge University Press.
- **Bhupatiraju, S.**, Chen, D., Jankin, S., Kim, G., Kupi, M., & Maqueda, M. R. (2023). Government Analytics Using Machine Learning. *The Government Analytics Handbook*.

Conference Proceedings

- Neis, P., **Bhupatiraju, S.**, Chen, D. L., & Joshi, S. (2024). Islamophobia in the Justice System and Judicial Mitigation in Bihar, India. In *18th Annual Conference on Empirical Legal Studies (CELS)*.

Working Papers

- **Bhupatiraju, S.**, Chen, D., Joshi, S., Neis, P., & Singh, S. (2024). Litigation as Scrutiny: A Four Decade Analysis of Environmental Justice, Firms, and Pollution in India.
- **Bhupatiraju, S.**, Chen, D. L., Das, R., Joshi, S., & Neis, P. (2022). Impact of free legal search on rule of law: Evidence from Indian Kanoon.
- **Bhupatiraju, S.**, Chen, D. L., Joshi, S., & Neis, P. (2021). Who Is in Justice? Caste, Religion and Gender in the Courts of Bihar over a Decade. *World Bank Policy Research Working Paper*.

SELECTED PROJECTS & APPLICATIONS

Witchcraft Court Case Pattern Analysis 2023–2024
Built multi-agent investigative system using CrewAI with specialized investigator personas; implemented HyDE for semantic search; deployed on Fly.io (witchcraft-cases.fly.dev).
Technologies: CrewAI, Atomic Agents, HyDE, Chroma

PFM Bottleneck Identification (Evidence Extraction) 2023–Present
Designed evidence extraction system for Public Financial Management; applied DSPy for automated prompt optimization; implemented reflection pattern for iterative quality improvement; conducted human-in-the-loop validation with domain experts.
Technologies: DSPy, LangChain, reflection patterns

Lok Adalat Communication Effectiveness Study (SAMA Partnership) 2022–2023
Designed A/B testing framework evaluating communication channel effectiveness for people's court notifications in India; findings informed SAMA's communication strategy (sama.live).
Technologies: A/B testing frameworks, statistical analysis

Personal Experimentation & Research Tools
Vector database exploration (Pinecone, Chroma, FAISS) for different retrieval scenarios; dynamic schema generation using FAISS for context-specific Pydantic enums in structured extraction; model optimization techniques including LoRA fine-tuning, knowledge distillation experiments, and quantization for local LLM deployment in resource-constrained environments.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

In Progress: Microsoft Azure Fundamentals (AZ-900)

Completed – DeepLearning.AI Courses

- Agentic AI
- Agentic Knowledge Graph Construction
- Pydantic for LLM Workflows
- DSPy: Build and Optimize Agentic Apps